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A framework for maximizing conservation stewardship effectiveness in mosaic landscapes: bringing together ecological priority and social feasibility

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Research Paper

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Summary

Effective conservation of rare species requires approaches that extend beyond protected areas and integrate the perspectives of diverse landowners. We synthesized results from rare plant assessments with landowner perceptions and conservation values in landscapes with a mix of ownership and management types to produce an integrated framework for maximizing stewardship effectiveness. We combined field surveys of rare plant species across five land-ownership types (private, provincial, federal, non-profit and leaseholder) with semi-structured interviews of representatives of these groups in the Redberry Lake Biosphere Region of Canada to identify conservation values, landowner attitudes, perceived challenges, preferred incentives and legislative awareness across ownership types. Derived from the results, our framework drew attention to ecological priority and social feasibility as key elements of integrated conservation management. The framework provided operational definitions for each element and highlighted that even regional conservation strategies must account for variation in landownership and values. This framework therefore serves as a cohesive approach to applying both ecological and social evidence to inform decision-making for biodiversity conservation in mosaic landscapes. Our experience suggests that bringing together evidence from the natural and social sciences offers novel insights, and the framework we propose may be transferable across a range of conservation contexts.

Introduction

Although Canada supports rich biodiversity and considerable ecological and economic wealth, a substantial number of species, including many vascular plants, are assessed as at risk (Ray et al. 2021, McCune et al. 2024), and current measures may be insufficient to secure their persistence. Canada has committed to protecting 30% of its land by 2030, yet existing protected areas cover only 12.4% of the country, which is insufficient to protect terrestrial biodiversity (ECCC 2023). Habitat loss has been identified as the largest threat to rare species in Canada, affecting over 82% of listed species (Woo-Durand et al. 2020), with agriculture (46%) and urbanization (44%) being the most common causes (Venter et al. 2006). Effectively addressing these challenges requires both robust biodiversity inventories, recognized as a foundational requirement for sustainable management and conservation (Wagensommer 2023), and evidence-informed policy reform (Buxton et al. 2021). It seems that integration of two key sources – empirical biodiversity data and perceptions and attitudes from multiple stakeholders – could be more valuable and have more impact than any of these sources on their own, because conservation actions are more likely to be adopted and to succeed when ecological priorities are matched to the social acceptability and feasibility of action on the ground (Whitehead et al. 2014).

Achieving this integration is particularly important in mosaic landscapes – where lands are owned and managed through a mix of ownership types and responsibilities – making understanding of landowner perceptions of conservation value essential for biodiversity protection. A key concern is that areas of high ecological value, where conservation efforts should be prioritized, do not always correspond with areas where landowner participation in conservation initiatives is expected to be high (Raymond & Brown 2011). This mismatch matters because acquiring and formally protecting all high-value land is rarely feasible; much of it must instead be conserved through the voluntary stewardship of the landowners who already hold and manage it. Since landowner management decisions directly influence rare species conservation outcomes, identifying the motivators and barriers to participation in conservation programmes is critical for designing effective initiatives (Sorice et al. 2013). Therefore, improved ecological

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knowledge of rare species, while necessary, is insufficient for biodiversity protection without a parallel understanding of landowner interest and engagement in conservation.

The Redberry Lake Biosphere Region (RLBR) in the Canadian province of Saskatchewan provides an excellent case study of a mosaic landscape, where competing land uses and diverse ownership types create both opportunities and obstacles for biodiversity protection. As the province's only biosphere reserve, the RLBR was designated by UNESCO in 2000 with a mandate encompassing conservation, sustainable development and capacity-building (RLBR n.d.). Situated within one of Canada's most intensive agricultural landscapes, the RLBR plays a critical role in safeguarding the future of regional biodiversity (Kricsfalusy 2026a). However, achieving this conservation mandate necessitates a systematic evaluation of rare species across diverse landownership types to guide targeted and effective stewardship.

The overarching goal of this study was to explore how decision-making in biodiversity conservation might be enhanced by integrating methods of the natural and social sciences. To ground the framework empirically, we focused on native flora, conducting rare plant assessment to document species and identifying their conservation status and distribution across landownership types in the region. We assessed social feasibility by conducting interviews with a range of landowners in the region. Combining these assessments, the framework places units according to their ecological priority and social feasibility, with each combination indicating a distinct set of management actions. Our results informed the efficacy of the framework and confirmed its value in identifying key elements of effective conservation stewardship.

Methods

Study area

The RLBR is located c. 100 km north-west of the city of Saskatoon (Saskatchewan, Canada). After expanding the territory of the Redberry Lake Biosphere Reserve in 2019 (RLBR n.d.) and committing to reconciliation with Indigenous Peoples, the area and corresponding organization were officially renamed the 'Redberry Lake Biosphere Region'. The RLBR is organized into three interrelated zones known as a core area, buffer zone and transition area, which are common for biosphere reserves. The saline Redberry Lake with several islands comprises the core area. The surrounding buffer zone is partially protected and has restrictions on allowable activities. The transition area comprises the largest portion and functions as an area for sustainable development where a variety of land uses are permitted, principally modern commercial agriculture (RLBR n.d.). More than 80% of the lands in the RLBR are either arable, under permanent crops or under pastures (Kricsfalusy 2023).

Establishing landownership groups and landownership mapping

We adopted broad landownership categories based on expert consultation, classifying lands according to both ownership and management type. Seven major landownership groups were identified: private, leaseholders, non-profit organizations, federal government, provincial government, rural municipalities and First Nations. The latter two groups were not represented among interview participants and were therefore excluded from analysis. Details of the five included groups are given in Table S1. Participants were identified through purposive sampling with help from the RLBR practitioners; contact details were obtained from publicly available online sources for institutional groups, and for private landowners they were shared by the RLBR staff with the landowner's prior permission. Institutional representatives (federal, provincial and non-profit) were individuals with direct on-the-ground management or stewardship responsibilities for the lands concerned.

To assess rare plant distribution across landownership, maps were created for the five ownership groups using a standardized GIS workflow in *ArcGIS Pro* (Esri 2022), with spatial data harmonized, clipped to the RLBR boundary and aggregated by ownership type. Landownership layers were compiled from provincial land administration and parks records, the Saskatchewan Digital Landcover (SDL n.d.) dataset, Prairie Farm Rehabilitation Administration (PFRA n.d.) pasture layers, federal Canadian Wildlife Service designations and non-profit property boundaries; where a parcel carried more than one designation, it was assigned to a single ownership category based on the primary managing authority, such that categories are mutually exclusive. Private lands combined farmland, North Cottage Area and Saskatoon Sailing Club areas, with cropland identified from the SDL and PFRA datasets. Leaseholder lands – mainly recreational cottage leases – were mapped from provincial records. The North Cottage Area consists of privately owned cottage lots and is therefore part of the private group, whereas leaseholder lands comprise cottage lots leased (not owned) within the Redberry Lake Regional Park. Provincial lands merged 13 categories (Table S1). Here, 'Pastures' refers to named provincial Crown (community) pastures administered by the Ministry of Agriculture and used primarily for managed livestock grazing. Non-profit lands combined conservation organization properties. Federal lands, including the Migratory Bird Sanctuary and National Wildlife Area, are both administered by the Canadian Wildlife Service and were treated as a single federal category; they were mapped as separate layers only to reflect their distinct legal designations.

Landowner interviews

Representatives from each landownership group (Table S1) participated in semi-structured interviews. Most participants were middle-aged (over 75% older than 45), well-educated (59% holding a bachelor's degree or higher) and predominantly male (65%; Table S2). The study followed the research ethics requirements of the University of Saskatchewan.

The semi-structured interview guide was organized into the following themes: landowner type; general perceptions of rare species; perceived conservation challenges; preferred incentives; awareness of relevant legislation; and demographics. Perceptions were elicited by asking participants how they would feel if a rare species occurred on their land, with six options (proud, excited, worried, annoyed, indifferent or other) and an invitation to elaborate; discussions addressed rare species generally, with particular emphasis on rare plants. For clarity, we distinguished three related concepts used throughout: awareness of rare species on one's property (knowing which occur and where); awareness of legislation (having heard of and being able to describe The Wildlife Act (TWA 1998) and the Species at Risk Act (SARA 2003)); and access to information (knowing where to obtain and being willing to seek guidance on rare species stewardship). In total, 17 individuals across 5 landownership groups participated in 13

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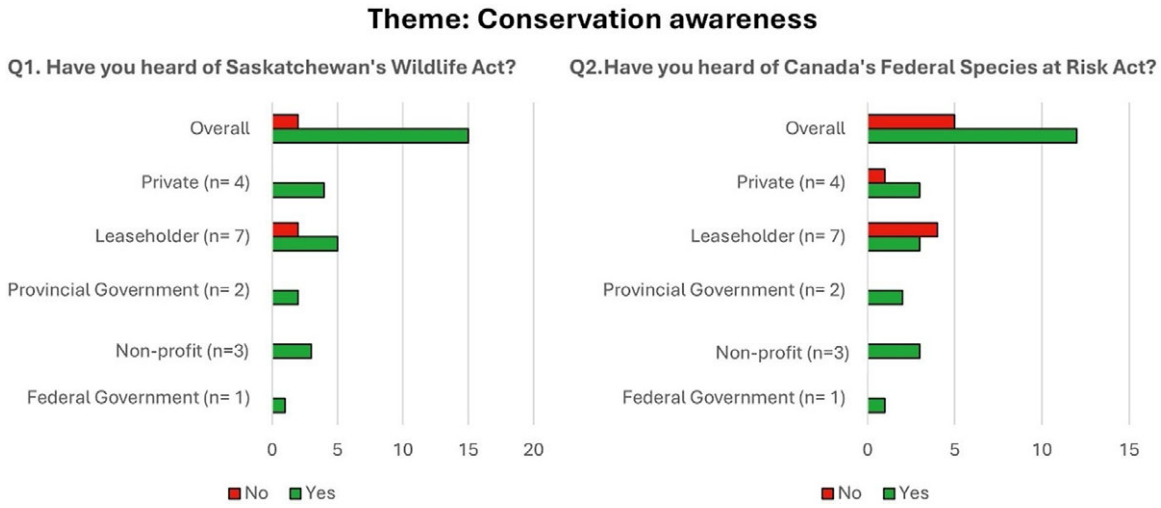


Figure 1. Responses on conservation awareness (whether participants had heard of each act; yes/no) by landownership group.

interviews conducted individually or in small groups via phone or email, lasting 15–45 min. All interviews were recorded, transcribed using Temi (2019) by hand and coded in *NVivo* 12.6 (QSR International 2019) using a combination of deductive coding (with codes defined a priori from the interview themes) and inductive coding (with codes emerging from the transcripts), supplemented by hand-coding. Exemplary quotes, cross-tabulation queries and frequency analyses were subsequently used to identify key perspectives and insights. Unless otherwise indicated, the percentages reported in the following sections are calculated from the full sample of 17 participants.

Rare plant assessment

Rare vascular plant assessment followed a targeted (purposive) design. It was not intended to represent or to sample in proportion to the area of each landownership type, and no ratio-based or statistical sampling design was applied; rather, its purpose was to document rare plants and provide their inventory. Surveys were conducted from 2010 to 2025, with routes covering all parts of the biosphere region to capture its full range of habitats. The detailed survey methodology is described in Kricsfalusy (2026b), and a comprehensive account of the distribution, richness and population attributes of the rare plants recorded is given in Kricsfalusy (2026a). Identified locations of rare plants were recorded using a GPS unit (Garmin Oregon 650t). Occurrence records were cross-checked against the Hunting, Angling and Biodiversity Information of Saskatchewan database (HABISask n.d.). Identified occurrence points were spatially intersected to calculate species richness, occurrence counts and density metrics (occurrences per km^2 and richness per km^2) for the five ownership categories with both mapped layers and survey coverage. Because survey effort was not proportional to the total area of each ownership type, per-area values are presented as descriptive summaries within ownership types and should not be interpreted as unbiased estimates of relative rare plant density among ownership types.

Plant specimens that could not be identified in the field were collected, pressed and later identified through voucher examination at the W.P. Fraser Herbarium (SASK n.d.), University of Saskatchewan. Species taxonomic nomenclature follows the Database of Vascular Plants of Canada or VASCAN (Brouillet et al. 2010+). Species conservation ranks were assigned following the Saskatchewan Conservation Data Centre (SKCDC n.d.).

Results

Specific elements emerged from our data collection that informed framework development.

How did landowners react to rare species occurrence?

The most common feelings about having a rare species were 'excited' (65%), 'proud' (41%) and 'worried' (41%). Negative responses included annoyance among private landowners (25%) and worry among leaseholders (71%); this worry was often directed not at regulation but at neighbouring land users, whom leaseholders felt they could not control if others acted destructively towards habitat. Additional concerns included the perception that hosting a rare species could impose financial hardship (e.g., through restrictions on productive land use) and a recurring emphasis on fairness in how conservation responsibilities are shared among landowners. Although the small and uneven sample (n = 17) limits firm conclusions about landownership effects, patterns suggest that non-profits expressed mostly positive or neutral views, private landowners were largely positive and leaseholders tended towards neutral or negative perspectives.

Participants were asked whether they had heard of Saskatchewan's TWA and Canada's SARA and, if so, to describe briefly what each governed (Fig. 1). Awareness was higher for the provincial law, with 88% of participants familiar with TWA, while 71% had heard of SARA. Lack of awareness was greater among leaseholders (57%) and private landowners (25%), whereas all government and non-profit representatives were familiar with both acts. Interestingly, when asked to elaborate, most leaseholders and private landowners associated TWA primarily with hunting, and several indicated they could 'probably guess' the purpose of SARA despite limited knowledge of its details.

All participants were aware of the RLBR, although one private landowner did not realize that their land was within the biosphere region. Most understood what rare species are, describing them as

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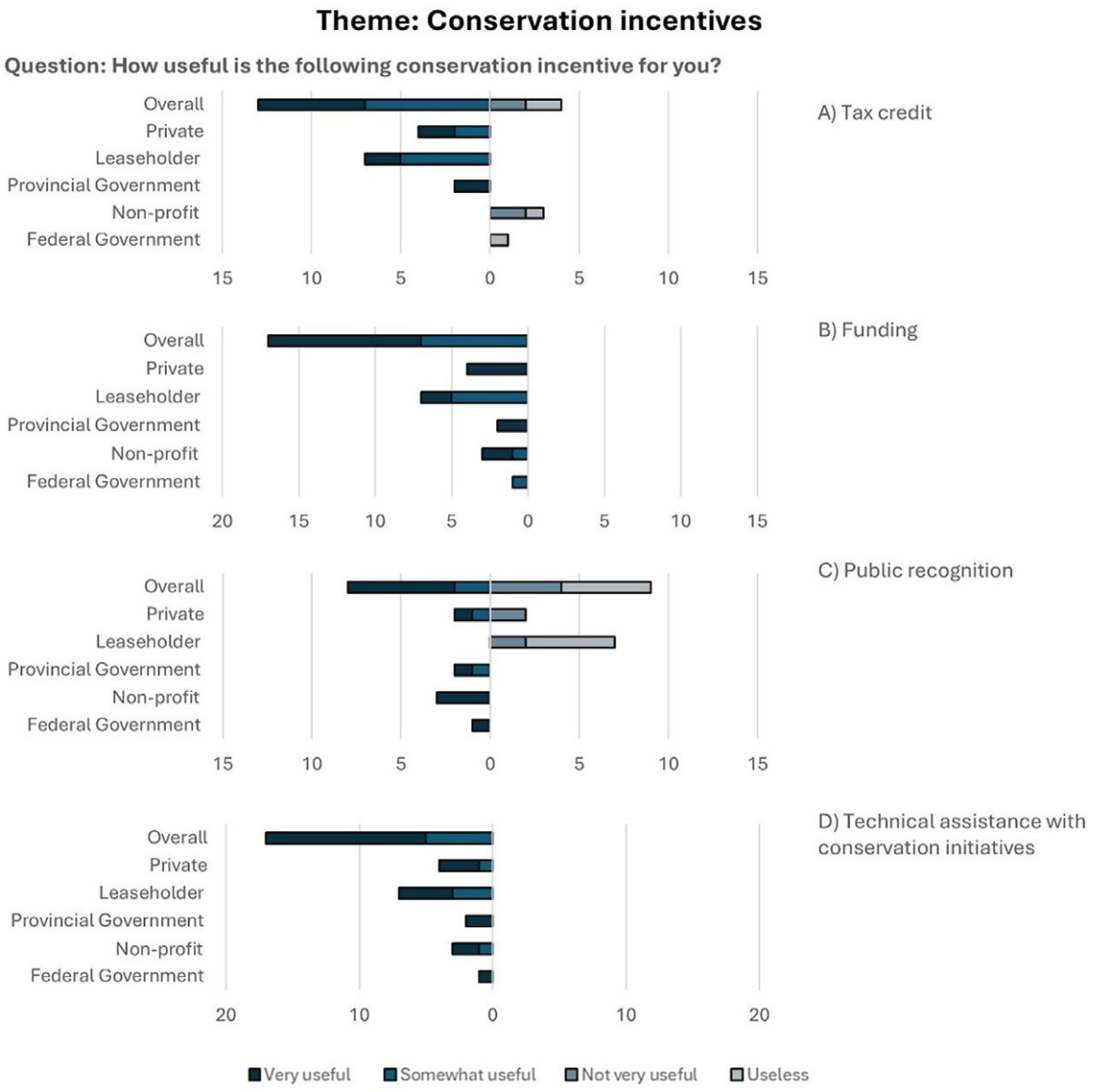


Figure 2. Responses on conservation incentives (usefulness on a four-point scale) by landownership group.

'close to extinction', 'struggling to reproduce' or simply 'never noticed before'. Knowledge of plants was limited across all landownership groups, and no participants expressed strong interest in or knowledge of rare plants.

Landowners ranked technical assistance (defined in interviews as expert advice and support from conservation professionals or agencies) as the most useful incentive, with 71% rating it very useful and none rating it as not useful (Fig. 2). One leaseholder noted that technical support was effective because it made conservation efforts 'easy'.

Representatives of the federal and provincial governments viewed tax credits (offered in exchange for participating in rare species stewardship) as irrelevant, as they pay no property tax, and non-profit representatives noted that they pay little or no property tax on most of their conservation lands. In contrast, half of private landowners and all leaseholders (71% somewhat useful, 29% very useful) considered them beneficial, although scepticism remained.

Representatives of the federal and provincial governments and of non-profit organizations viewed public recognition as somewhat or very useful for enhancing public image. In contrast, all leaseholders rated it as not very useful or useless, and most private landowners felt it was not personally motivating but could persuade others.

What were the perceived conservation challenges and constraints?

The main conservation challenges were related to species awareness (Fig. 3). Most participants (58%) did not know which rare species occurred on their property, and 53% did not know their locations, with all leaseholders reporting this as a significant challenge; these figures reflect participants' self-reported knowledge of rare species in general, as interviews and plant surveys were not matched parcel by parcel. In contrast, when asked whether it was important to know which rare species occurred on their land,

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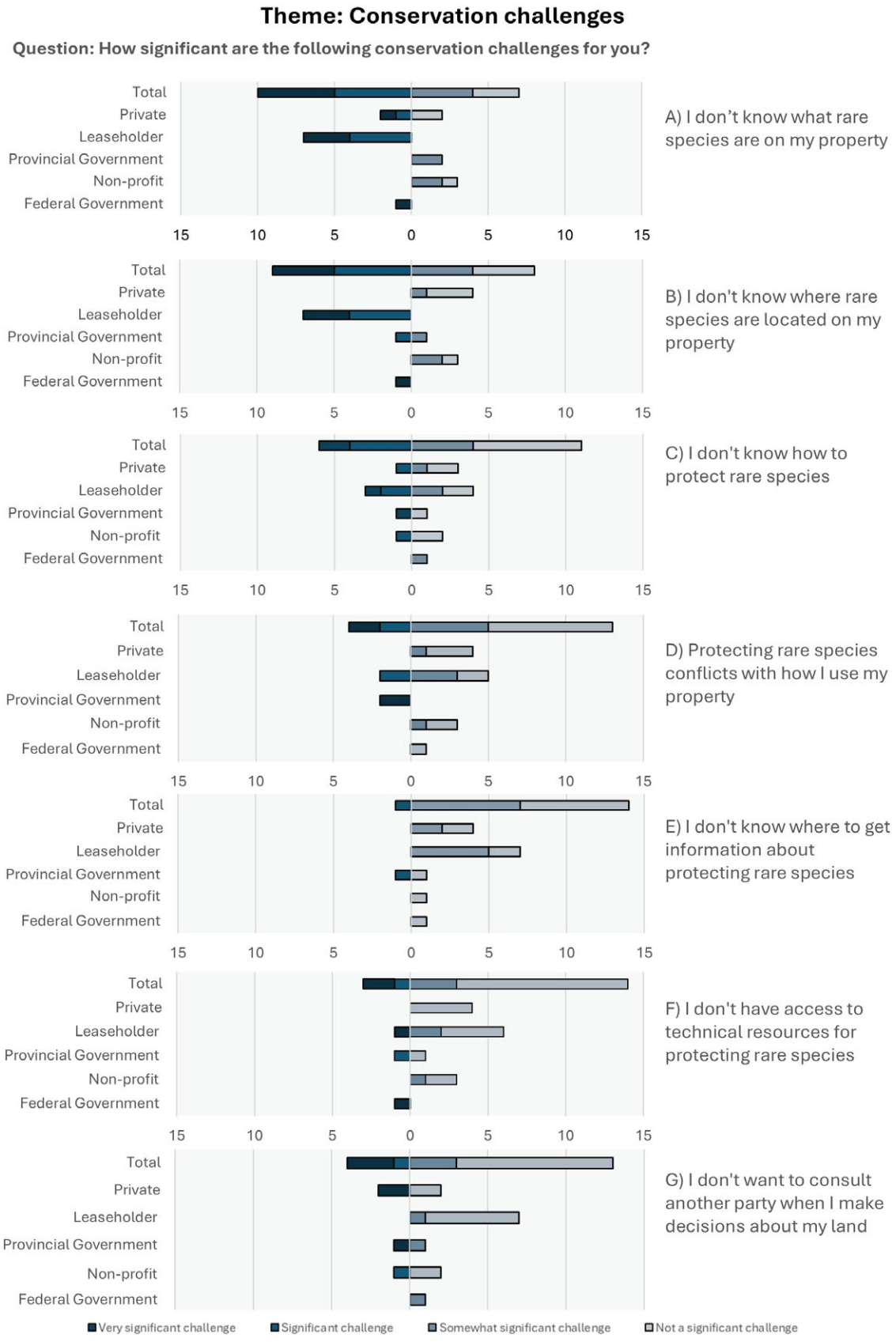


Figure 3. Responses to questions about the significance of various conservation challenges by landownership group. Colours represent a four-point Likert scale.

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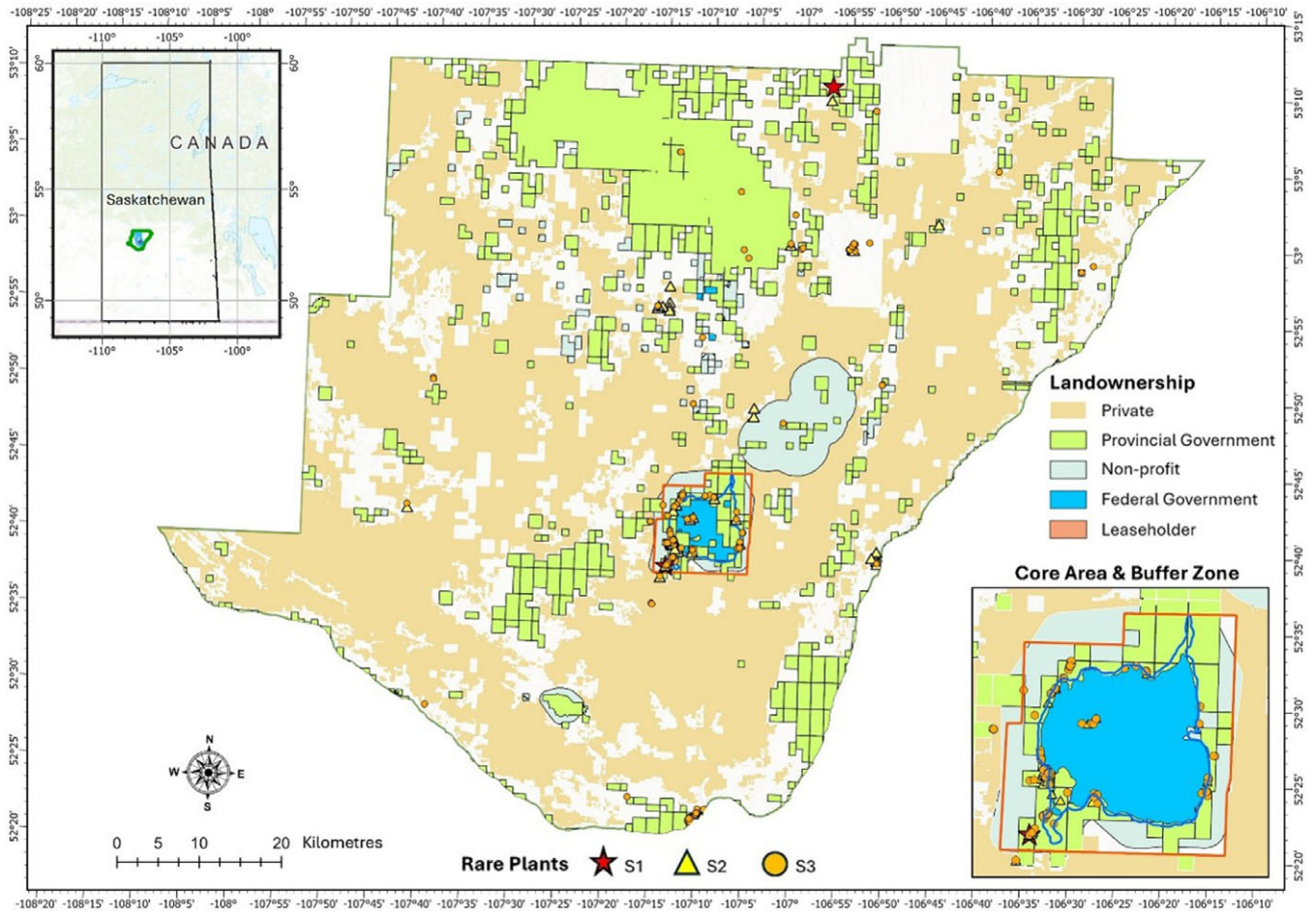


Figure 4. Landownership patterns and rare plant species distribution in the Redberry Lake Biosphere Region. Landownership: dark blue parcels are federal lands (not water); white or unshaded areas are lands outside the five mapped ownership categories. Species conservation ranks: S1 = Critically Imperilled; S2 = Imperilled; S3 = Vulnerable.

half of private landowners did not consider this a significant challenge, and 75% said that they were confident about the locations of rare species on their properties.

Some 41% of participants felt that protecting rare species was not a significant challenge, although their level of confidence varied. Some 50% of private landowners agreed with this perspective; however, only 29% of leaseholders felt that protecting rare species would not be a significant challenge. Finding information online was generally easy for landowners, but several leaseholders indicated that they would not seek it out proactively; as one put it, '[W]e don't seek this information until it's provided for us,' pointing to the need for direct, proactive outreach.

Information emerged as the main conservation challenge, although most participants (65%), including all private landowners and 57% of leaseholders, did not regard limited access to technical resources for protecting rare species as a significant challenge. Few participants (12%) anticipated land-use conflicts, although all provincial representatives perceived them as significant; 75% of private landowners reported no expected conflicts (Fig. 3).

Participants were also asked how challenging it would be to have to consult another party (e.g., land manager or agency) when making decisions about their land. Overall, 59% did not consider this a significant challenge, reflecting most leaseholders (86%), who are already required to consult the regional park authority. Half of private landowners considered consultation particularly challenging, and both federal and provincial representatives noted difficulties in coordinating among governments, rural municipalities and landowners; as one federal representative remarked, '[B]etter relationships … [and] cooperation for conservation would still be beneficial.' This suggests that working relationships among agencies and landowners exist but are not yet strong enough to coordinate rare species management across ownership boundaries.

How were rare plants distributed across landownership groups?

Field surveys identified 46 vascular plants of conservation concern, defined as those assigned a rank from Critically Imperilled to Vulnerable by the SKCDC (n.d.), representing 8.1% of all rare species in Saskatchewan. A comprehensive account of their distribution, richness and population attributes is provided in a separate publication (Kricsfalusy 2026a). Among these plants of conservation concern at the global and national levels, five species were assessed as Vulnerable (2 G3 and 3 N3). At the subnational (provincial) level, however, the degree of concern was greater, with 37 species classified as S3 (Vulnerable), 7 as S2 (Imperilled) and 2 as S1 (Critically Imperilled; Table S3). In terms of landownership, private landholders manage the largest share of the RLBR (55.7%), followed by the provincial government (35.0%) and non-profit organizations (8.6%); the federal government and leaseholders together comprise less than 1.0% of lands (Fig. 4 & Table S4).

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Private lands were predominantly agricultural, comprising croplands, followed by the North Cottage Area and the Saskatoon Sailing Club. Across these lands, 26 occurrences of 10 rare plant species were recorded, with low density reflecting the extensive agricultural areas. These occurrences were in uncultivated features embedded within the cropland matrix (field margins, sloughs and remnant native grassland patches) rather than in tilled ground; as one private landowner observed, '[T]here're only a few places they can be in, and that's not where we're ploughing the ground.' Croplands accounted for most occurrences (24 occurrences of 10 species: 2 S2 and 8 S3), while the Saskatoon Sailing Club supported two species (1 S2 and 1 S3) with two occurrences. No rare species were recorded in the North Cottage Area.

Leaseholders' land comprised cottagers at the Redberry Lake Cottage Area who were leasing land for leisure purposes in the Redberry Lake Regional Park. We did not find any rare plant species occurrences in leaseholder lands.

Provincial lands encompassed 13 land types and supported 34 rare plant species (2 S1, 5 S2 and 27 S3). Agricultural Crown Lands covered the largest area, recording the highest species richness (26 species: 2 S1, 3 S2 and 21 S3) and 86 occurrences, followed by Wildlife Habitat Protection Areas with 16 species (3 S2 and 13 S3) and 76 occurrences. Regional Parks contained nine species (1 S2 and 8 S3), while Royal Pasture supported three S3 species. Historic Parks and Recreation Sites each recorded a single S2 species, and Wildlife Refuge Lands contained two species (1 S2 and 1 S3). No rare species were recorded in remaining pastures, Game Preserves or Crown Conservation Easements.

Non-profit lands supported 216 occurrences of 23 rare plant species (1 S1, 3 S2 and 19 S3). Among these, Important Bird Areas (IBAs) recorded the highest number of occurrences (157) and species richness (19 species: 1 S1, 2 S2 and 16 S3). Redberry Lake Regional Park had the highest rare species density per unit area within this group, with 40 occurrences of 9 species (1 S2 and 8 S3), followed by Redberry Lake Bible Camp, with 14 occurrences of 5 species (1 S2 and 4 S3). Fish and Wildlife Development Fund lands contained two species (1 S2 and 1 S3), and a single S3 species was recorded on Saskatchewan Wildlife Federation lands. No rare species occurrences were recorded on Ducks Unlimited Canada or RLBR-owned lands.

Federal lands comprised a Migratory Bird Sanctuary and a National Wildlife Area. The former accounted for the larger share of federal lands and all nine occurrences of the four recorded rare plant species, while the National Wildlife Area covered a smaller area with no rare species occurrences. Of the four species recorded, one is rated S2 and three are S3.

Discussion

Landowner conservation values and motivations

Overall, participants viewed the presence of rare species on their own land positively, consistent with Olive and McCune (2017). However, where Ontario landowners' appreciation of nature coexisted with resistance to regulation, RLBR private landowners, although wary of consultation, mostly expected no land-use conflict. The main negative feeling was the worry expressed by government representatives and leaseholders that others might harm rare species, and that expressed by leaseholders that hosting them might restrict their use of the property. Such local perceptions matter because they shape whether landowners support, tolerate or resist conservation measures on their land (e.g., whether they permit surveys, adopt recommended management or enrol in stewardship programmes), and thus they carry social, ecological and governance consequences (Bennett 2016). Positive attitudes often reflect environmental values, whereas negative attitudes are more often driven by perceived financial risk and concerns over property rights (Vizek & Nielsen-Pincus 2017). Feelings of pride and excitement may therefore support conservation engagement, although concerns about fairness in sharing conservation responsibilities were also noted.

The most valued conservation incentive was technical assistance from conservation experts, which is consistent with findings that landowners value expert guidance (Underwood et al. 2012). Funding ranked second, although some leaseholders and private landowners expressed distrust of 'free money'; this reflected suspicion about hidden costs and where the funds come from, as participants asked whether they were simply paying for it through their own taxes. This indicates that funding is most effective when paired with transparency about its source and conditions and with explicit assurances against future land-use restrictions (Langpap 2004). In Saskatchewan, voluntary conservation easements (CEA 1996) can be structured to allow flexibility and, in some cases, withdrawal, which helps address concerns about permanent loss of autonomy (Government of Saskatchewan n.d.).

Public recognition was valued by non-profits and governments but seen as ineffective by most leaseholders and private landowners. Given the low awareness of rare species in the RLBR, low-cost educational tools (e.g., brochures with images of locally rare plants) and peer networks of landowners may be practical means of raising awareness and encouraging beneficial management, complementing rather than replacing formal incentives (Brown 2019, Rust et al. 2022).

Challenges and constraints to conservation

The main challenge across landowners was not knowing which rare species were on their property, with even federal representatives citing limited time, funds and labour for inventories. A recent Canada-wide survey of practitioners indicated habitat loss, weak enforcement of protection laws and low public and landowner awareness as major barriers to the conservation and recovery of at-risk plants (McCune et al. 2024). Because many landowners are unlikely to seek information independently, outreach should deliver it directly, such as through targeted mailings, species lists and identification brochures, site visits and messages routed via trusted non-environmental networks (e.g., rural municipality associations and the regional park board).

Participants were less aware of and less interested in rare plants than in birds. Rare plants were sometimes perceived as nuisances or 'weeds', contrasting with more favourable views of birds, a finding that is consistent with Brook et al. (2003). Awareness of legislation was limited: most had heard of SARA and TWA but could not describe their contents, echoing findings from Henderson et al. (2014). Landowners cannot protect what they do not recognize, and they are unlikely to act on obligations that they are unaware of, which identifies awareness-raising and plant-focused education as priorities for conservation programming in the RLBR.

Stewardship of rare species

Private lands, mostly cropland in the transition zone, are where agriculture threatens rare plant persistence primarily by converting

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and fragmenting native habitat and, at the margins, through disturbance into the uncultivated remnants where these plants occur (Kricsfalusy 2023, 2026a,b). Despite these pressures, private agricultural lands remain important for conservation, as they support several rare species. Across Canada, approximately a third of at-risk plant species are documented primarily on private lands (McCune & Morrison 2020, McCune et al. 2024).

Interviews indicate that private landowners hold positive attitudes towards conservation but limited knowledge of species and policy, and many are wary of external land-use regulation, echoing challenges in maximizing participation in voluntary conservation programmes reported elsewhere in Canada (Reiter et al. 2022). Because provincial legislation protects species but not habitats and imposes no duties on private owners, voluntary stewardship programmes are especially valuable here; such programmes have worked in the region, including Saskatchewan's Rare Plant Rescue (Vinge-Mazer & Ranalli 2012) and Alberta's MULTISAR (MULTISAR n.d.), offering transferable models for the RLBR.

Leaseholder lands form a small part of the RLBR and had no recorded rare species, reflecting their limited area and intensive recreational use. Leaseholders expressed neutral to negative attitudes, tied to specific concerns: they must already consult the park authority, they were worried that rare species designations could further restrict recreation. and they felt unable to control neighbouring users. Engaging them will therefore depend less on new obligations than on reassurance, simplicity and a clear demonstration of benefit.

Provincial lands are large and diverse, spanning all three zones. They shield four of seven rare plant-richness micro-hotspots (Kricsfalusy 2026a) and hold the highest overall richness, and their representatives reported strong positive attitudes. Because these holdings span grazed Crown pastures, recreation sites and protected areas, rare plant protection can be built into existing multi-use management.

Non-profit lands supported 23 rare species and lie within or near the core and buffer zones. This probably reflects the fact that non-governmental organizations preferentially acquire parcels of known conservation value and manage them to maintain rare plant habitat, although outcomes depend on the legal, economic and institutional frameworks guiding management (Pasquini et al. 2011). Supporting and expanding non-profit acquisition and stewardship through funding, partnerships and secure tenure could therefore be one of the most effective levers for protecting rare plants.

Federal lands support the highest rare species density of any ownership category, despite being designated primarily for birds and other wildlife (Government of Saskatchewan n.d.). Their small extent limits overall richness, but the presence of rare species shows that federal designations can provide important rare plant habitat where conditions remain protected.

Government programmes that relied on the threat of regulation rather than incentives and collaboration have often bred distrust (Langpap 2004). Private lands therefore benefit from awareness and stewardship programmes and government lands from formal protection, monitoring and restoration, with targeted acquisition complementing, not replacing, voluntary stewardship. On Agricultural Crown Lands, results-based approaches, tax incentives, extension and adjusted grazing-lease conditions can promote multi-species conservation (Ayambire et al. 2021).

Our research suggests that putting these strategies into practice requires active engagement between incentive providers, including the Government of Saskatchewan, Ducks Unlimited Canada, the Fish and Wildlife Development Fund and land managers. Effective strategies should combine protection, including acquisition where warranted, of high-value parcels with outreach, technical assistance and incentives that engage all landowners, and they can be strengthened by novel tools such as Important Plant Areas and other effective conservation measures (see also Kricsfalusy 2026a, Kricsfalusy et al. 2026).

The contribution of the framework

We produced a novel framework for integrated conservation management that brings together ecological priority with social feasibility to inform an integrative understanding of conservation and to point to key actions that influence its effectiveness (Fig. 5). A key insight is that even within the same ecological region, the areas where species are rare or at risk may not be those where landowners are willing or able to act; hence, applying the same strategy everywhere is inefficient and ineffective.

The RLBR is a clear case with five different landownerships, where the regional flora is exceptionally rich, comprising 584 native taxa, of which 46 are rare (Kricsfalusy 2026b). Much of this threatened flora, however, lies outside the protected core and buffer zones, on agricultural lands where cultivation and land-use pressure are high (Kricsfalusy 2026a).

By applying the framework, we can see provincial lands gain greater scope to build habitat protection (Fig. 5). The Landowner Stewardship Perspective Analysis indicates high social feasibility: provincial lands, as the second-largest ownership group after private lands, give any initiative wide reach; their representatives reported strong positive attitudes towards conservation; and, because the land is held under the Crown, few acquisition barriers stand in the way. The Rare Species Assessment also indicates high ecological priority, as most rare species on provincial lands occur on Agricultural Crown Land, more than half of which is unprotected in the buffer zone of the RLBR. These lands have already been proposed as an Important Plant Area to protect the Redberry Lake Flats site, which supports 12 threatened plant species and sensitive habitats (Kricsfalusy et al. 2026). Such protection can be reinforced by proper designation, technical assistance and incentive programmes for Crown Land managers and by outreach and recognition to sustain their engagement.

Figure 5 links the two evidence streams to integrated management actions. Conservation responses should vary with ecological priority and social feasibility (Fig. 6): high priority × high feasibility supports direct protection and management; high priority × low feasibility calls for building feasibility through education, technical assistance and incentives; low priority × high feasibility supports ecological assessment, restoration and voluntary stewardship; and low priority × low feasibility requires maintaining awareness and periodically reassessing ecological status and social feasibility to inform future management actions. Table S5 makes the decision logic explicit by summarizing the conservation actions that each combination implies. By evaluating ecological priority and social feasibility together for each type rather than in parallel, the framework moves beyond simply establishing that ecological and social evidence both matter to specifying decision-orientated tools for biodiversity conservation in mosaic landscapes.

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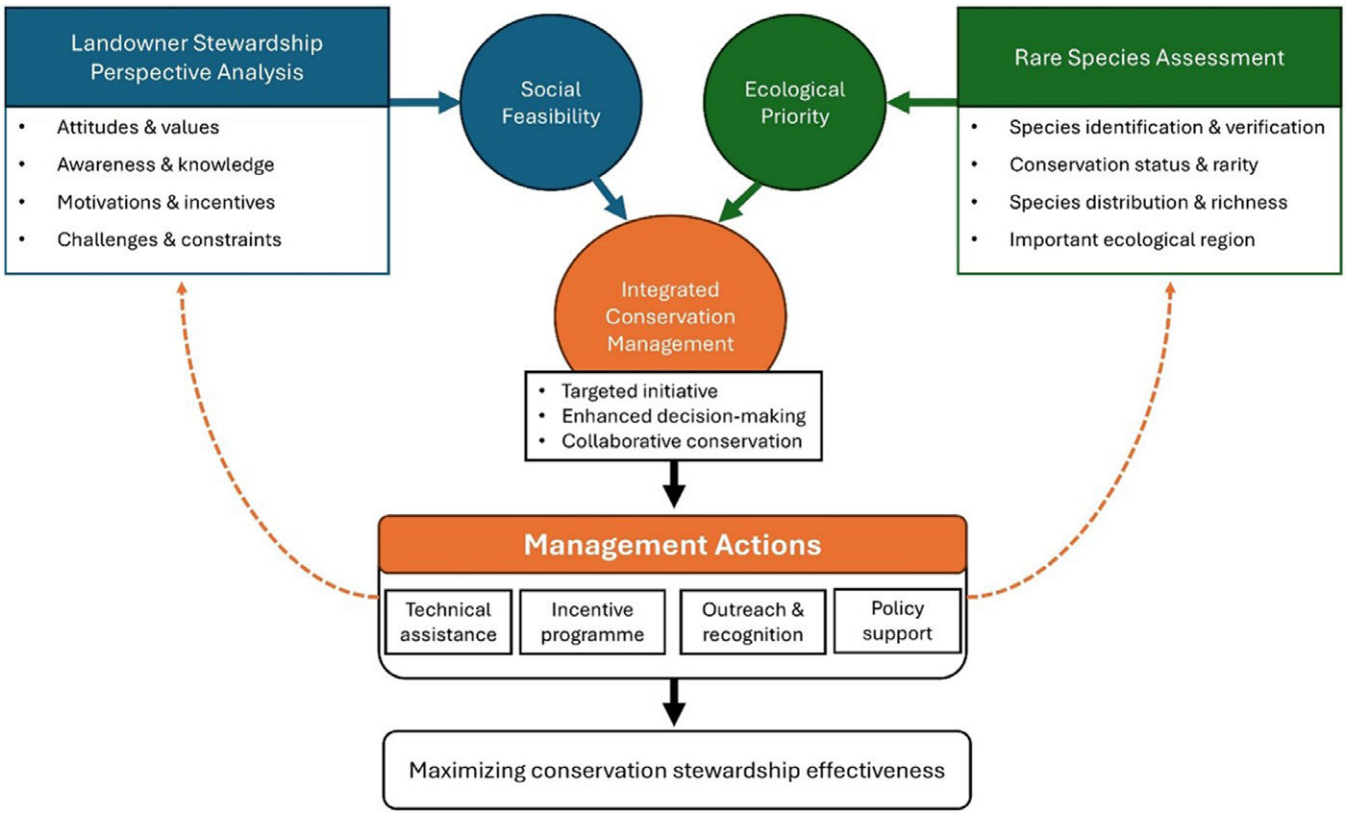


Figure 5. Integrated framework for maximizing stewardship effectiveness in mosaic landscapes.

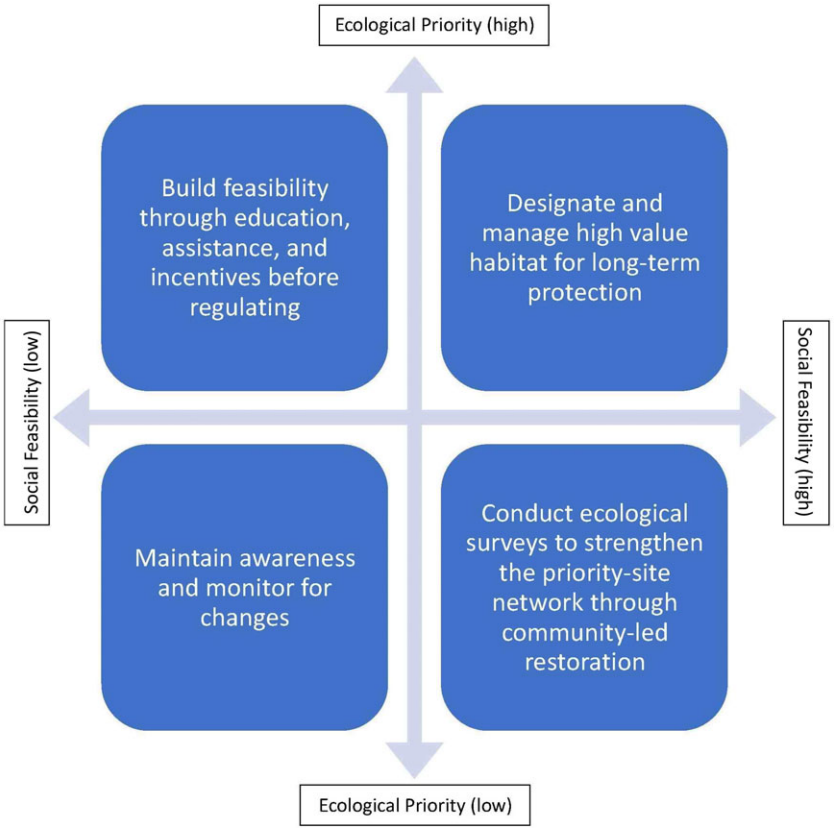


Figure 6. Management actions across typologies of ecological priority and social feasibility.

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Conclusion

A key contribution of this study is that it brings to bear both methods of the natural and social sciences on the understanding of the potential and challenges of biodiversity conservation in mosaic landscapes. By doing so, we reveal not only the unique characteristics of rare plant species, but also the perspectives of landowners on specific barriers to conservation programming. Because these barriers differed by ownership type, areas within the same ecological region may require different conservation strategies. Through a novel approach that integrates ecological priority with social feasibility, this research provides a strong foundation to support informed decision-making and advance biodiversity conservation in the study region, and it offers transferable lessons for conservation practice elsewhere.

Supplementary material. To view supplementary material for this article, please visit https://doi.org/10.1017/S0376892926100617.

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